PRASAD RAO

QA Engineer

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PROFESSIONAL SUMMARY

- 10+ years of QA Automation Engineering experience with strong expertise in Selenium WebDriver, Cypress, and Playwright — building scalable, maintainable automation frameworks using JavaScript and TypeScript with POM, Data-Driven, and Hybrid design patterns in Agile/Scrum environments.
- Built AI agents for QA at Comcast using GitHub Copilot integrated with the client's MCP server: one reads Cucumber and Playwright test reports to pinpoint why each test failed and recommend the fix, and another generates test scenarios from user stories and acceptance criteria.
- Proven track record in functional testing, regression testing, smoke testing, browser compatibility testing, and end-to-end automation across web, mobile, API, and media/streaming application platforms.
- Hands-on experience testing media and streaming applications — validated video playback pipelines, adaptive bitrate (ABR) streaming, HLS/DASH stream integrity, OTT platform workflows, CDN content delivery, DRM license validation, and media metadata accuracy for telecom and entertainment platforms.
- Expert in Cypress (JavaScript/TypeScript) for modern web application automation — dynamic UI behavior, API mocking, network interception, full E2E user journey validation, and CI/CD pipeline integration via Jenkins and GitHub Actions.
- Deep Playwright (TypeScript) expertise — cross-browser testing (Chromium, Firefox, WebKit), network interception, AI-assisted self-healing locators, and intelligent retry mechanisms delivering 40% faster regression execution and 60% reduction in flaky tests.
- Strong API testing proficiency using Postman and REST Assured — validating REST/SOAP/GraphQL endpoints, JSON/XML payloads, headers, and status codes across microservices integrations with full version control via Git/GitHub.
- Experienced with JIRA and TestRail for test management — test plans, execution cycles, defect lifecycle tracking, traceability matrices, and Agile sprint QA reporting to product owners and release managers.
- Performance testing expertise using JMeter with cloud testing on AWS and Azure — load, stress, and endurance testing for high-volume streaming and web APIs; validated SLA compliance and throughput under peak traffic conditions.
- BDD practitioner — authored Cucumber Gherkin feature files implemented with Selenium-Cucumber and Cypress-Cucumber frameworks integrated with Jenkins CI/CD for business-aligned acceptance testing.
- Mobile automation specialist using Appium on Android and iOS — validated streaming, media playback, and application workflows across devices with ADB log capture for root cause analysis.
- ISTQB Certified Tester (Foundation Level) | Lambda Test Certified Professional — strong analytical skills and attention to detail with a proven track record of embedding quality into every stage of the Agile delivery lifecycle.

TECHNICAL SKILLS

Core Automation	Selenium WebDriver (Java/JavaScript), Cypress (JavaScript/TypeScript), Playwright (TypeScript/JavaScript) — POM, Data-Driven, Hybrid Frameworks
Languages	JavaScript (Primary), TypeScript (Primary), Java, Python, SQL, VBScript
API Testing	Postman, REST Assured, REST/SOAP/GraphQL, JSON/XML Payload Validation, Network Interception, API Mocking & Stub Services
Media & Streaming QA	Video streaming pipeline testing, Adaptive bitrate (ABR) validation, HLS/DASH stream verification, OTT platform testing, CDN content delivery, DRM license validation, Playback & buffering analysis
BDD Frameworks	Cucumber (Gherkin/BDD), Jasmine, Mocha, Jest — Business-aligned acceptance test automation integrated with CI/CD
Mobile Automation	Appium (Android/iOS), Xcode, Android Studio, ADB log analysis, Cross-platform mobile testing

Performance Testing	JMeter (Load/Stress/Endurance), Azure Load Testing, New Relic APM, Grafana, SLA & Throughput Validation
Test Management	JIRA, TestRail, Zephyr Scale, Xray, Confluence, HP ALM/Quality Center, Defect Lifecycle Management
CI/CD & Version Control	Jenkins, GitHub Actions, Azure DevOps, Git, GitHub, GitLab, Bitbucket, Maven, Docker, Kubernetes
Cloud Testing	AWS (EC2, S3, Lambda, RDS), Azure, GCP, Cloud-based browser testing, Microservices & Containerized environments
Browser Compatibility	Chrome, Firefox, Safari, Edge, WebKit — Selenium Grid, Playwright multi-browser, Browser Stack, Cross-device testing
Monitoring & Observability	Splunk, New Relic, Datadog, Azure Monitor, Application Insights
Databases	Oracle, SQL Server, PostgreSQL, MongoDB — Advanced SQL, Table Joins, Stored Procedures, ETL Validation
SDLC / Methodologies	Agile (Scrum), DevOps, Waterfall, STLC, V&V, SQA Best Practices, Sprint Planning, Retrospectives

PROFESSIONAL EXPERIENCE

QA Engineer

Comcast | Philadelphia, PA | *April 2026 – Present*

- Built an AI-powered test failure analysis agent that reads Cucumber and Playwright test reports, pinpoints the exact reason each test failed, and recommends the fix, separating genuine product defects from environment and script issues to cut manual triage time.
- Developed an AI-driven test case generation solution that creates test scenarios from user stories and acceptance criteria, accelerating test design and widening coverage of edge cases.
- Created an automated comparison script to validate localized content between SC8 and SC10 during platform migration, flagging missing, mismatched, or untranslated strings before release.
- Owned QA for the launch of the OTP (one-time passcode) authentication flow, automating tests in Playwright to validate code delivery, expiry, retry limits, invalid-code handling, and resend behavior; the feature shipped to production with no post-release defects.
- Caught a critical defect in a new Okta authentication release where duplicate cookies and session storage entries were created for a single user, which could have caused session conflicts and login issues for customers; logged it with evidence so it was fixed before reaching production.
- Executed Playwright test suites through Jenkins pipelines and reviewed run results to catch regressions before release.
- Tested REST APIs using Bruno, validating request and response payloads, status codes, and error handling.
- Coordinated test data provisioning with upstream teams and shared ready-to-use data sets across the QA team to keep sprint testing unblocked.
- Mentored and guided a team of 6 QA engineers on Playwright automation standards, test design, and defect triage, and reviewed their test scripts for quality and consistency.
- Coordinated daily between onshore and offshore QA teams, handing off test priorities, blockers, and defect status across time zones to keep sprint delivery on schedule.
- Analyzed application logs in Kibana to investigate test failures and defects, tracing errors to their root cause to speed up triage and resolution.
- Monitored daily order reports to track order fallout, investigated why orders failed, and shared the findings with the respective teams for resolution.
- Supported the Services and Workbench teams during their releases, performing production validation and providing QA sign-off, including night and weekend deployments.
- Executed functional, regression, and smoke testing each sprint, logged and tracked defects in JIRA, and took part in sprint planning, stand-ups, and retrospectives.

Technology Stack: *Python/JavaScript, Playwright, Cucumber, GitHub Copilot, MCP, Okta, Kibana, Bruno, JIRA, Git, Jenkins*

QA Automation Engineer

Verizon | (Remote) | *February 2025 – March 2026*

- Designed and built scalable QA automation frameworks using Selenium WebDriver (JavaScript), Cypress (TypeScript), and Playwright (TypeScript) for Verizon's media streaming and OTT platform — validated video playback pipelines, adaptive bitrate (ABR) streaming, HLS/DASH stream integrity, CDN content delivery, and DRM license validation across web and mobile surfaces.
- Developed and maintained Cypress (TypeScript) automation suites covering full E2E streaming user journeys — content discovery, search and filters, video player launch, bitrate switching, buffering behavior, closed captions, login/authentication, subscription management, and payment workflows.
- Built Playwright (TypeScript) cross-browser automation across Chromium, Firefox, WebKit, and Edge — implemented network interception to assert CDN request/response integrity, media segment loading, and API-driven content metadata accuracy.
- Implemented AI-assisted testing techniques within Playwright frameworks — self-healing locator strategies, AI-driven test prioritization for high-risk streaming scenarios, and intelligent retry mechanisms reducing flaky test failures by 60% and regression execution time by 40%.
- Executed API testing using Postman and REST Assured — validated REST/GraphQL endpoints for content catalog, user entitlements, streaming session tokens, playback URLs, and media metadata; asserted JSON payloads, response headers, and status codes across microservices.
- Integrated Cypress and Playwright test suites into Jenkins and GitHub Actions CI/CD pipelines — enforced automated quality gates, nightly regression runs, parallel cross-browser execution, and defect trend dashboards for engineering and product leadership.
- Performed SQL-based backend validation using Oracle and PostgreSQL queries with table joins — verified content metadata, user entitlement records, and streaming session data integrity between API responses and database records.
- Mentored 5+ QA engineers in JavaScript/TypeScript automation best practices, Cypress and Playwright framework standards, Git version control workflows, and sprint QA readiness criteria.

Technology Stack: *Cypress (TypeScript), Playwright (TypeScript), Selenium WebDriver (JavaScript), Postman, REST Assured, Cucumber BDD, JMeter, Jenkins, GitHub Actions, Git, JIRA, TestRail, Zephyr Scale, Oracle, PostgreSQL, AWS, Appium, New Relic, Splunk, Selenium Grid, HLS/DASH, OTT/CDN, DRM*

QA Automation Engineer

5Y Business Solutions (Clients: RoyFi, LDV, Epic Will, Health Rise) | Remote — Franklin, TN | *March 2022 – December 2024*

- RoyFi (fintech): Tested a platform that provides credit to music artists based on their popularity, validating credit and repayment workflows; caught a defect where a referral account received its discount on every repayment instead of only one, which would have caused ongoing revenue loss.
- Epic Will (legal tech): Tested a will management application covering all 50 states, validating the end-to-end flow in which families create their wills, attorneys review them, and the final documents are securely stored.
- LDV (rental marketplace): Tested a platform for renting beach chairs, umbrellas, boats, and other equipment across multiple US locations, covering search, availability, booking, and payment flows.
- Health Rise (health insurance): Tested patient claims processing and validated incoming claims data for accuracy and completeness.
- Built scalable Selenium WebDriver (Java/JavaScript) automation frameworks from scratch using POM, Hybrid, and Data-Driven design patterns with TestNG and Maven — delivered E2E functional test coverage for fintech, legal-tech, rental booking, and health insurance applications in Agile Scrum environments.
- Developed Cypress (JavaScript) automation suites for LDV’s rental booking engine — covered search, filters, availability modules, booking flows, payment processing, date restrictions, blackout periods, refund rules, and guest info validation across dynamic UI behaviors.
- Executed hybrid UI + API automation workflows using Cypress and Postman — validated that frontend user actions correctly triggered backend API services and database updates, catching integration defects 3 sprints earlier than previous manual testing approach.
- Performed comprehensive API testing using Postman and REST Assured — validated REST endpoint request/response payloads in JSON and XML formats; performed mock API testing for payment processors and third-party integrations using Postman stub services.
- Authored BDD test scenarios in Gherkin and implemented with Cucumber-Selenium and Cucumber-Cypress frameworks — produced reusable feature files integrated with Jenkins CI/CD pipelines for business-aligned acceptance testing.

- Managed test execution, defect tracking, and traceability using JIRA with Xray and Zephyr Scale — maintained bidirectional traceability from business requirements to test cases and defect records across all sprint releases.
- Configured Selenium Grid and Jenkins pipelines for parallel, cross-browser test execution of Epic Will across Chrome, Firefox, and Edge — reduced regression runtime through distributed parallel test execution.
- Implemented CI/CD-driven regression pipelines using GitHub Actions and Jenkins — enabled continuous testing across development, staging, and UAT environments with automated quality gate enforcement.
- Performed SQL-based backend validation using Oracle and PostgreSQL queries with table joins — verified booking, billing, invoicing, and delivery data integrity across multi-source backend systems.
- Executed JMeter load and stress tests for LDV's booking and payment APIs and Epic Will's will-creation APIs; used Azure Load Testing for cloud-based performance validation on AWS and Azure — measured response time, throughput, error rates, and SLA adherence under peak traffic.
- Conducted Appium mobile automation on Android and iOS for the LDV and Epic Will apps — validated application workflows, UI responsiveness, and functional regression across device types using Xcode and Android Studio.
- Monitored application health and API performance for Epic Will and RoyFi using New Relic APM and Azure Monitor — correlated transaction traces and error rates with QA test execution for faster root cause analysis.
- Led offshore QA team members in test execution coordination, defect resolution, and sprint QA velocity — ensured consistent quality process adherence across time zones using JIRA and Confluence.

Technology Stack: *Selenium WebDriver (Java/JS), Cypress (JavaScript), Playwright, Postman, REST Assured, Cucumber BDD, JMeter, Azure Load Testing, Jenkins, GitHub Actions, Git, JIRA, Zephyr Scale, Xray, TestRail, Oracle, PostgreSQL, AWS, Azure, Appium, New Relic, Confluence*

QA Automation Tester

Helen of Troy | El Paso, TX | *July 2018 – February 2022*

- Designed and built Selenium WebDriver (Java/JavaScript) automation frameworks using POM, TestNG, Maven, and Log4J for E2E functional and regression test coverage of a multi-channel eCommerce platform — validated catalog, cart, checkout, payment gateways, promotional engines, coupon logic, and order management workflows.
- Built and maintained Cypress (JavaScript/TypeScript) automation suites for UI regression and smoke testing of React-based storefronts — integrated into Azure DevOps CI/CD pipelines for continuous regression coverage across desktop and mobile web.
- Automated 90% of REST API tests using REST Assured and Postman — reduced API validation time by 30% and ensured 99% response accuracy across payment processor APIs including authorization, capture, refund, and chargeback flows.
- Executed browser compatibility testing across Chrome, Firefox, Safari, and Edge using Selenium Grid — validated consistent UI rendering and functional behavior across all supported browser environments.
- Authored Cucumber BDD feature files in Gherkin — implemented Selenium-Cucumber frameworks with Jenkins pipeline integration for business-aligned acceptance testing and sprint release validation.
- Wrote advanced SQL queries with table joins across Oracle and SQL Server for backend validation of order, billing, and inventory records ensuring data integrity across distributed system integrations.
- Executed JMeter load and stress tests for critical checkout and payment APIs — measured throughput, response time, and error rates under peak seasonal traffic and validated SLA compliance.
- Performed Appium mobile regression testing on Android and iOS — achieved 80% test coverage and reduced mobile regression execution time by 20%.
- Managed test execution and defect tracking in JIRA — communicated defect trends, test coverage metrics, and release readiness to stakeholders; issued UAT and production deployment signoffs.

Technology Stack: *Selenium WebDriver (Java/JS), Cypress (JavaScript/TypeScript), Postman, REST Assured, Cucumber BDD, JMeter, Jenkins, Azure DevOps, Git, TestNG, Maven, Oracle, SQL Server, Docker, Appium, JIRA, Selenium Grid*

Software Test Engineer

Vensai Technologies Pvt. Ltd. | Hyderabad, India | *June 2016 – July 2018*

- Performed system, integration, GUI, functional, and regression testing for healthcare and enterprise web applications — validated patient registration, appointment scheduling, clinical data entry, and insurance claim processing workflows.

- Created data-driven automated test cases in QTP/UFT using VBScript and descriptive programming — automated eligibility checks, provider lookup, and benefit verification regression scenarios.
- Executed smoke and feature-based regression tests across multiple builds; logged and tracked defects in HP Quality Center (ALM) and Bugzilla with full traceability documentation.
- Wrote SQL queries against SQL Server databases — verified data inserts, updates, and stored procedure outputs for functional and integration test validations.
- Mapped test cases to requirements ensuring 100% end-to-end coverage; maintained traceability matrices linking test cases to originating user stories and business requirements.
- Performed HIPAA compliance testing — validated PHI data masking, role-based access controls, and user authentication security controls for healthcare application modules.
- Provided onboarding support to new QA team members; acted as liaison between onsite and offshore teams for issue clarification, test scope alignment, and delivery coordination.

Technology Stack: *QTP/UFT, VBScript, Manual Testing, SQL Server, HP Quality Center (ALM), Bugzilla, ASP.NET, MS Office, TOAD, Windows*

EDUCATION & CERTIFICATIONS

B. Tech – Electrical & Electronics Engineering | JNTU, Hyderabad, India | 76%

- ISTQB Certified Tester — Foundation Level (CTFL)
- Lambda Test Certified Professional — Test Automation
- Claude Code 101 from Anthropic
- Claude Code in Action, Anthropic
- Introduction to Model Context Protocol (MCP), Anthropic (Claude Academy), Oct 2026
- Playwright & Agentic AI: Building Autonomous Test Frameworks